Causal Survival - Case Base

2026-05-28

Imports

EDA

```r
library("tableone")
library("xtable")
```

Cox PH Model

```r
library("survival")
```

SMD Balance

```r
library("survey")
library("cobalt")
```

Case Base

```r
library("casebase")
```

EDA

Data Set

```
#set seed
set.seed(2026)

#re-factor the data set
valid.indices <- seq(2, nrow(colon), by = 2)

#main variables of interest
time <- colon$time[valid.indices]
Delta <- colon$status[valid.indices]
A <- ifelse(
  colon$rx[valid.indices] == "Obs",
  0,
  1
)

#covriates
Gender <- colon$sex[valid.indices]
Age <- colon$age[valid.indices]
Obstruction <- colon$obstruct[valid.indices]
Perforation <- colon$perfor[valid.indices]
Adherence <- colon$adhere[valid.indices]
NumNodes <- colon$nodes[valid.indices]
Differentiation <- colon$differ[valid.indices]
Extent <- colon$extent[valid.indices]
Surgery <- colon$surg[valid.indices]
Nodes4 <- colon$node4[valid.indices]

#data set design
data.set <- data.frame(
  time = time,
  Delta = Delta,
  A = A,
  Gender = Gender,
  Age = Age,
  Obstruction = Obstruction,
  Perforation = Perforation,
  Adherence = Adherence,
  NumNodes = NumNodes,
  Differentiation = Differentiation,
  Extent = Extent,
  Surgery = Surgery,
  Nodes4 = Nodes4
)
```

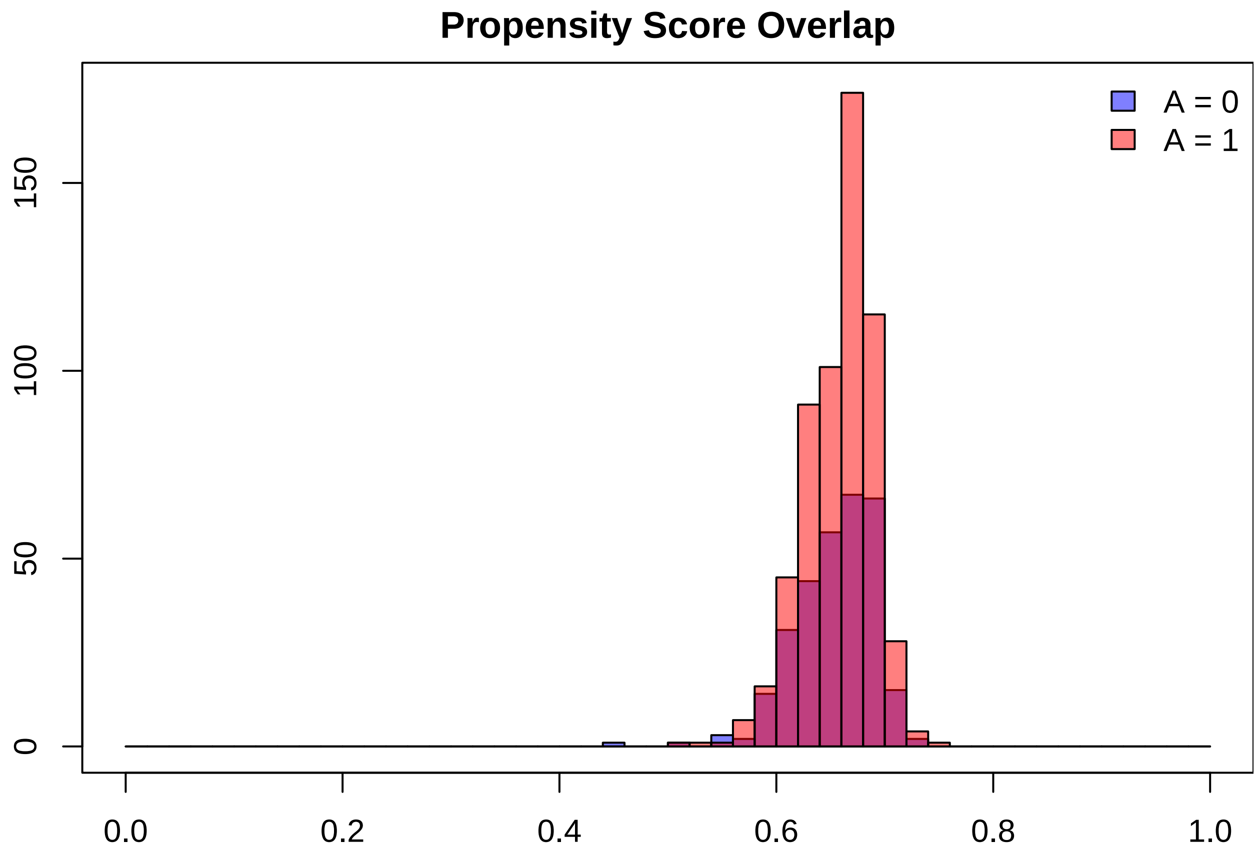
Table One

```r
covariates = c(
  "Gender",
  "Age",
  "Obstruction",
  "Perforation",
  "Adherence",
  "NumNodes",
  "Differentiation",
  "Extent",
  "Surgery",
  "Nodes4"
)

table.one.original <- CreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set,
  test = FALSE
)

print(table.one.original, smd = TRUE)
```

```
##                               Stratified by A
##                                0             1             SMD
##   n                              315           614
##   Gender (mean (SD))            0.53 (0.50)   0.52 (0.50)   0.018
##   Age (mean (SD))              59.45 (11.97) 59.91 (11.94)  0.038
##   Obstruction (mean (SD))       0.20 (0.40)   0.19 (0.39)   0.024
##   Perforation (mean (SD))       0.03 (0.17)   0.03 (0.17)   0.004
##   Adherence (mean (SD))         0.15 (0.36)   0.14 (0.35)   0.017
##   NumNodes (mean (SD))          3.79 (3.73)   3.59 (3.49)   0.053
##   Differentiation (mean (SD))  2.08 (0.50)   2.05 (0.52)   0.054
##   Extent (mean (SD))            2.89 (0.52)   2.88 (0.47)   0.015
##   Surgery (mean (SD))           0.29 (0.45)   0.25 (0.44)   0.078
##   Nodes4 (mean (SD))            0.28 (0.45)   0.27 (0.45)   0.006
```

Propensity Scores

```
#generate propensity scores
propensity.score <- fitted(
  glm(
    A ~ Gender +
      Age +
      Obstruction +
      Perforation +
      Adherence +
      NumNodes +
      Differentiation +
      Extent +
      Surgery +
      Nodes4,
    data = data.set,
    family = binomial(link = "logit")
  )
)

#separate propensity scores for exposed & unexposed groups
propensity.score.0 <- propensity.score[A == 0]
propensity.score.1 <- propensity.score[A == 1]

#visualize the propensity scores overlap
par(
  mar=c(2,2,2,0)
)

hist(
  propensity.score.0,
  col=rgb(0,0,1,0.5),
  breaks=seq(0,1,by=0.02),
  ylim=c(0,175),
  main="Propensity Score Overlap",
  xlab="Propensity Scores"
)

hist(
  propensity.score.1,
  col=rgb(1,0,0,0.5),
  breaks=seq(0,1,by=0.02),
  add=T
)

box()

legend(
  "topright",
  legend=c('A = 0', 'A = 1'),
  fill=c(rgb(0,0,1,0.5),rgb(1,0,0,0.5)),
  border="black", bty="n"
)
```


Propensity Score Overlap
A = 0
A = 1
0
50
100
150
0.0
0.2
0.4
0.6
0.8
1.0

Stabilized IPW

```r
p.A1 <- mean(A)

#calculate stabilized ipw
ipw <- ifelse(
  A == 1,
  p.A1 / propensity.score,
  (1 - p.A1) / (1 - propensity.score)
)
```

Covariate Balance

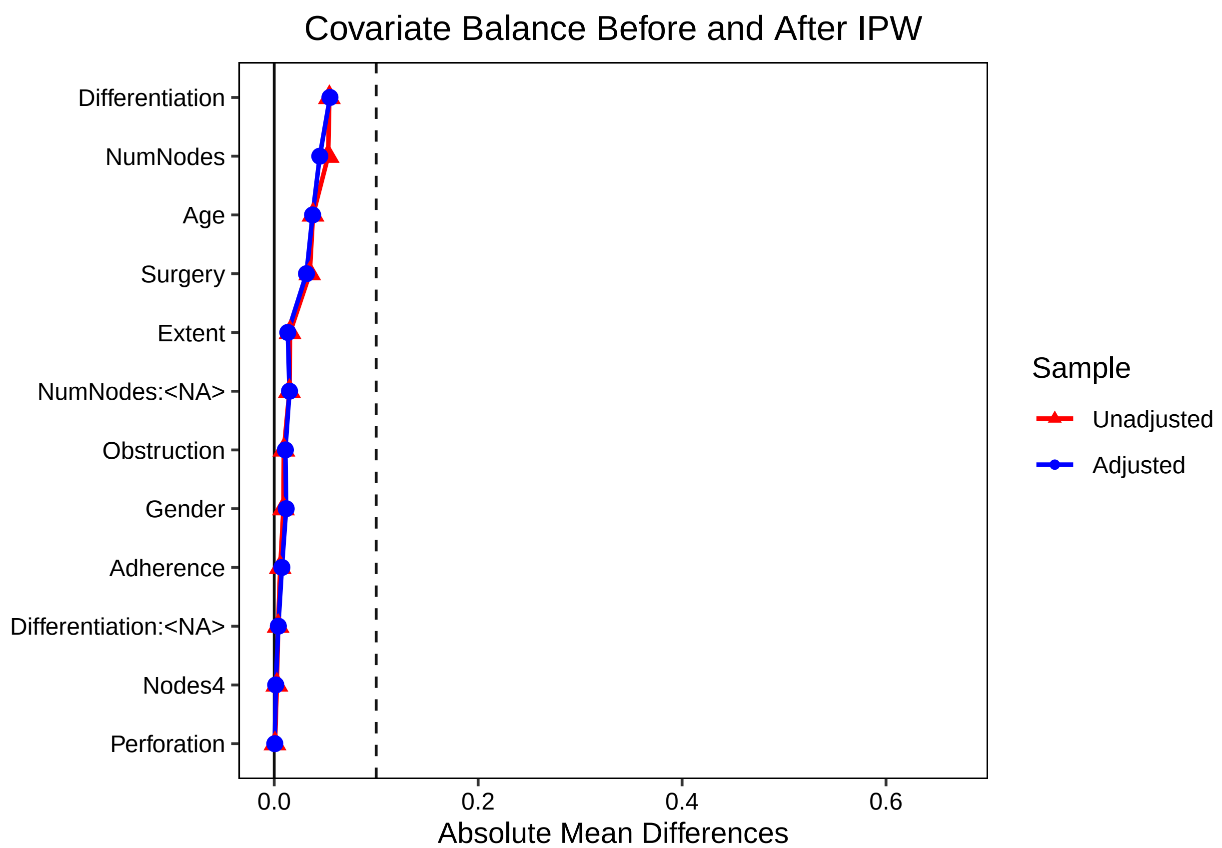
```
#calculate the covariate balance before & after IPW
balance <- bal.tab(
  A ~ Gender +
    Age +
    Obstruction +
    Perforation +
    Adherence +
    NumNodes +
    Differentiation +
    Extent +
    Surgery +
    Nodes4,
  data = data.set,
  weights = ipw,
  method = "weighting",
  estimand = "Average Treatment Effect",
  un = TRUE
)
```

```
## Warning: Missing values exist in the covariates. Displayed values omit these
## observations.
```

```
## Note: `s.d.denom` not specified; assuming "pooled".
```

```
#visualize the covariate balance before & after IPW
love.plot(
  balance,
  stats = "mean.diffs",
  threshold = 0.1,
  abs = TRUE,
  line = TRUE,
  var.order = "unadjusted",
  colors = c("red", "blue"),
  shapes = c(17, 16),
  limits = c(m = c(0, 0.7)), #hard-coded 0.7??
  title = "Covariate Balance Before and After IPW",
  labels = TRUE
)
```

```
## Warning: Standardized mean differences and raw mean differences are present in the same
## plot. Use the `stars` argument to distinguish between them and appropriately
## label the x-axis. See `love.plot()` for details.
```



Covariate Balance Before and After IPW
Differentiation
NumNodes
Age
Surgery
Extent
NumNodes:<NA>
Obstruction
Gender
Adherence
Differentiation:<NA>
Nodes4
Perforation
0.0
0.2
0.4
0.6
Absolute Mean Differences
Sample
Unadjusted
Adjusted

Table Two

```r
data.set.ipw <- svydesign(
  ids = ~ 0,
  data = data.set,
  weights = ipw
)

table.one.ipw <- svyCreateTableOne(
  vars = covariates,
  strata = "A",
  data = data.set.ipw,
  test = FALSE
)

print(table.one.ipw, smd = TRUE)
```

```
##                              Stratified by A
##                               0              1              SMD
##   n                           312.24         619.11
##   Gender (mean (SD))            0.53 (0.50)    0.52 (0.50)   0.023
##   Age (mean (SD))              59.49 (11.96)  59.94 (11.92)  0.038
##   Obstruction (mean (SD))       0.20 (0.40)    0.19 (0.39)   0.027
##   Perforation (mean (SD))       0.03 (0.17)    0.03 (0.17)   0.003
##   Adherence (mean (SD))         0.15 (0.36)    0.14 (0.35)   0.021
##   NumNodes (mean (SD))          3.76 (3.66)    3.60 (3.50)   0.045
##   Differentiation (mean (SD))   2.08 (0.50)    2.05 (0.52)   0.055
##   Extent (mean (SD))            2.89 (0.52)    2.89 (0.47)   0.013
##   Surgery (mean (SD))           0.29 (0.45)    0.26 (0.44)   0.071
##   Nodes4 (mean (SD))            0.28 (0.45)    0.27 (0.45)   0.003
```

Estimation

Assumptions

Non-Informative Censoring

Model 1: Unadjusted Cox

```r
#fit model
model.1 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  robust = TRUE
)

#generate summary
summary(model.1)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, robust = TRUE)
##
##   n= 929, number of events= 468
##
##       coef exp(coef) se(coef) robust se     z Pr(>|z|)
## A -0.24847   0.77999  0.09536   0.09522 -2.61  0.00907 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##   exp(coef) exp(-coef) lower .95 upper .95
## A      0.78      1.282    0.6472      0.94
##
## Concordance= 0.528  (se = 0.011 )
## Likelihood ratio test= 6.64  on 1 df,   p=0.01
## Wald test            = 6.81  on 1 df,   p=0.009
## Score (logrank) test = 6.82  on 1 df,   p=0.009,   Robust = 6.61  p=0.01
##
##   (Note: the likelihood ratio and score tests assume independence of
##      observations within a cluster, the Wald and robust score tests do not).
```

Model 2: Covariate Adjusted Cox

```r
#fit model
model.2 <- coxph(
  Surv(time, Delta) ~ A +
    Gender +
    Age +
    Obstruction +
    Perforation +
    Adherence +
    NumNodes +
    Differentiation +
    Extent +
    Surgery +
    Nodes4,
  data = data.set,
  robust = TRUE
)

#generate summary
summary(model.2)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A + Gender + Age + Obstruction +
##     Perforation + Adherence + NumNodes + Differentiation + Extent +
##     Surgery + Nodes4, data = data.set, robust = TRUE)
##
##   n= 888, number of events= 446
##    (41 observations deleted due to missingness)
##
##                      coef exp(coef)  se(coef) robust se      z Pr(>|z|)
## A               -0.250517  0.778398  0.098111  0.100421 -2.495 0.012608 *
## Gender          -0.117794  0.888879  0.095628  0.098407 -1.197 0.231300
## Age             -0.003005  0.996999  0.004039  0.004290 -0.701 0.483541
## Obstruction      0.184189  1.202243  0.119212  0.129646  1.421 0.155402
## Perforation      0.203951  1.226238  0.257374  0.263943  0.773 0.439696
## Adherence        0.175332  1.191641  0.130537  0.137084  1.279 0.200895
## NumNodes         0.042312  1.043220  0.015273  0.014828  2.854 0.004323 **
## Differentiation  0.122099  1.129866  0.098189  0.106769  1.144 0.252798
## Extent           0.449393  1.567360  0.118270  0.124649  3.605 0.000312 ***
## Surgery          0.243083  1.275175  0.104090  0.105027  2.314 0.020642 *
## Nodes4           0.568994  1.766489  0.142338  0.149080  3.817 0.000135 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##                 exp(coef) exp(-coef) lower .95 upper .95
## A                  0.7784     1.2847    0.6393    0.9477
## Gender             0.8889     1.1250    0.7330    1.0780
## Age                0.9970     1.0030    0.9887    1.0054
## Obstruction        1.2022     0.8318    0.9325    1.5501
## Perforation        1.2262     0.8155    0.7310    2.0570
## Adherence          1.1916     0.8392    0.9109    1.5589
## NumNodes           1.0432     0.9586    1.0133    1.0740
```

```
## Differentiation    1.1299     0.8851    0.9165    1.3929
## Extent             1.5674     0.6380    1.2276    2.0011
## Surgery            1.2752     0.7842    1.0379    1.5666
## Nodes4             1.7665     0.5661    1.3189    2.3660
##
## Concordance= 0.66  (se = 0.013 )
## Likelihood ratio test= 121.7  on 11 df,   p=<2e-16
## Wald test            = 121.9  on 11 df,   p=<2e-16
## Score (logrank) test = 137.2  on 11 df,   p=<2e-16,   Robust = 106.5  p=<2e-16
##
##   (Note: the likelihood ratio and score tests assume independence of
##      observations within a cluster, the Wald and robust score tests do not).
```

Model 3: Marginal Structural Cox

```r
#fit model
model.3 <- coxph(
  Surv(time, Delta) ~ A,
  data = data.set,
  weights = ipw,
  robust = TRUE
)

#generate summary
summary(model.3)
```

```
## Call:
## coxph(formula = Surv(time, Delta) ~ A, data = data.set, weights = ipw,
##     robust = TRUE)
##
##   n= 929, number of events= 468
##
##       coef exp(coef) se(coef) robust se      z Pr(>|z|)
## A -0.25405   0.77566  0.09525   0.09517 -2.669   0.0076 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##   exp(coef) exp(-coef) lower .95 upper .95
## A    0.7757      1.289    0.6437    0.9347
##
## Concordance= 0.529  (se = 0.011 )
## Likelihood ratio test= 6.95  on 1 df,   p=0.008
## Wald test            = 7.13  on 1 df,   p=0.008
## Score (logrank) test = 7.15  on 1 df,   p=0.007,   Robust = 6.92  p=0.009
##
##   (Note: the likelihood ratio and score tests assume independence of
##      observations within a cluster, the Wald and robust score tests do not).
```

Model 4: IPW Case Base PH

```r
#fit model
model.4 <- fitSmoothHazard(
  Delta ~ time * A,
  data = data.set,
  ratio = 10,
  weights = ipw
)
```

```
## 'time' will be used as the time variable
```

```r
#generate summary
summary(model.4)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 929
## Number of events: 468
## Number of base moments: 4680
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ time * A, data = data.set,
##     ratio = 10, weights = ipw)
##
## Coefficients:
##               Estimate Std. Error z value Pr(>|z|)
## (Intercept) -6.830e+00  1.214e-01 -56.236  < 2e-16 ***
## time        -1.213e-03  1.548e-04  -7.836 4.67e-15 ***
## A           -2.481e-01  1.525e-01  -1.627    0.104
## time:A      -1.304e-05  1.938e-04  -0.067    0.946
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3136.5  on 5147  degrees of freedom
## Residual deviance: 2900.8  on 5144  degrees of freedom
## AIC: 2908.8
##
## Number of Fisher Scoring iterations: 6
```

Model 5: IPW Case Base (Variable Hazard over Time)

```r
#fit model
model.5 <- fitSmoothHazard(
  Delta ~ pspline(time, df = 2) * A,
  data = data.set,
  ratio = 10,
  weights = ipw
)
```

```
## 'time' will be used as the time variable
```

```
## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
```

```r
#generate summary
summary(model.5)
```

```
## Fitting smooth hazards with case-base sampling
##
## Sample size: 929
## Number of events: 468
## Number of base moments: 4680
## ----
##
## Call:
## fitSmoothHazard(formula = Delta ~ pspline(time, df = 2) * A,
##     data = data.set, ratio = 10, weights = ipw)
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## (Intercept)                -21.2678     4.4278  -4.803 1.56e-06 ***
## pspline(time, df = 2)1      16.6195     5.1833   3.206  0.00134 **
## pspline(time, df = 2)2      12.8732     4.0649   3.167  0.00154 **
## pspline(time, df = 2)3      13.1538     4.7300   2.781  0.00542 **
## pspline(time, df = 2)4      12.0228     4.2617   2.821  0.00479 **
## pspline(time, df = 2)5      11.1893     5.2189   2.144  0.03203 *
## pspline(time, df = 2)6      17.2239     8.5903   2.005  0.04496 *
## pspline(time, df = 2)7     -93.7193   311.5750  -0.301  0.76357
## A                            2.8798     5.7397   0.502  0.61585
## pspline(time, df = 2)1:A    -3.7211     6.7085  -0.555  0.57911
## pspline(time, df = 2)2:A    -2.4710     5.2699  -0.469  0.63914
## pspline(time, df = 2)3:A    -3.9404     6.1606  -0.640  0.52242
## pspline(time, df = 2)4:A    -3.1935     5.4914  -0.582  0.56087
## pspline(time, df = 2)5:A     0.1437     7.3667   0.020  0.98443
## pspline(time, df = 2)6:A -105.2600    72.9599  -1.443  0.14910
## pspline(time, df = 2)7:A  456.9375  2199.2513   0.208  0.83541
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 3136.5  on 5147  degrees of freedom
```

```
## Residual deviance: 2842.3  on 5132  degrees of freedom
## AIC: 2874.3
##
## Number of Fisher Scoring iterations: 14
```